

Arava Kaushal Rao

(909) 352-8378 | kaushalarava7@gmail.com | [linkedin.com/in/kaushal-arava-rao-622b667b](https://www.linkedin.com/in/kaushal-arava-rao-622b667b) | github.com/kaushalarava | www.akrofficial.com

SUMMARY

Senior Full Stack Engineer, Product Architect, and Telematics Specialist with 10+ years of experience designing, developing, and scaling enterprise software platforms across transportation technology, fleet telematics, enterprise communications, and real-time collaboration systems. Proven expertise in Angular, React, TypeScript, Node.js, Java, Python, REST APIs, Microservices, SQL, Electron, WebRTC, SIP, and WebSockets. Experienced in building Transportation Management Systems (TMS), Fleet Telematics Platforms, Enterprise Communication Solutions, and Real-Time Messaging Systems. Strong background in product architecture, system design, technical leadership, customer-facing implementations, IoT ecosystems, operational support, and end-to-end software development lifecycle management.

In addition to enterprise software development, I have hands-on experience building AI-powered applications using OpenAI APIs and Large Language Models (LLMs).

Key Products: Anytrek Fleet Platform, ENSYNC Transportation Management System (TMS), MultiLine Desktop (T-Mobile), Roombr Video Conferencing Platform, Tale of Treasures AI Storytelling Platform

EDUCATION

California State University, San Bernardino, CA, USA

Master of Science (M.S.), Information Systems and Technology, GPA: 3.70/4.0

Graduated: May 2026

Sathyabama University, Chennai, Tamil Nadu, India

Bachelor of Science (B.S.), Information Technology, GPA: 7.9/10.0

Graduated: March 2014

EXPERIENCE

TELEMATICS IMPLEMENTATION SPECIALIST

March 2026 – Present

Anytrek Corporation

Ontario, CA, USA

Product: Anytrek Fleet & Telematics Platform

- Support the development and implementation of fleet management and telematics solutions for transportation customers.
- Design and implement frontend applications, backend services, REST APIs, and database integrations supporting telematics data processing and fleet visibility.
- Configure and integrate telematics devices, GPS tracking systems, vehicle sensors, and third-party fleet management platforms.
- Integrate GPS tracking devices, telematics hardware, and vehicle sensors to support real-time fleet visibility, operational analytics, and device telemetry monitoring.
- Develop dashboards, analytics, and reporting modules providing insights into fleet performance, maintenance schedules, route efficiency, fuel consumption, compliance metrics, and operational trends.
- Collaborate with customers to understand operational requirements and deliver tailored telematics and fleet management solutions.
- Support customer onboarding, implementation planning, user training, solution adoption initiatives, and post-implementation support.
- Partner with sales and business development teams to provide technical demonstrations, solution consulting, customer presentations, and product recommendations.
- Gather customer feedback and business requirements, translating them into platform enhancements and product improvements.

- **Technologies:** JavaScript, TypeScript, SQL, REST APIs, GPS Tracking, Device Telemetry, Fleet Management Systems, Telematics, IoT Solutions, Reporting & Analytics

PRODUCT ENGINEER & TRANSPORTATION OPERATIONS MANAGER

November 2023 – November 2025

Connect America Transportation Inc.

Fontana, CA, USA

Product: ENSYNC – Transportation Management System (TMS)

- Architected and developed ENSYNC, a Transportation Management System supporting dispatch management, route planning, shipment tracking, fleet operations, and logistics workflows.
- Designed scalable application modules supporting load management, customer management, driver assignments, route optimization, and transportation operations.
- Built responsive web applications enabling dispatchers and operations teams to efficiently manage transportation workflows.
- Developed backend services and RESTful APIs supporting shipment processing, dispatch scheduling, customer management, reporting, and operational analytics.
- Managed transportation operations including dispatch coordination, load planning, driver management, shipment monitoring, customer communication, and ELD-related operational workflows.
- Served as a subject matter expert for transportation operations, ELD compliance processes, fleet compliance requirements, dispatch management, and logistics workflows.
- Investigated operational incidents and customer-reported issues by analyzing transportation data, workflows, and platform behavior to identify root causes and implement corrective actions.
- Collaborated with dispatchers, drivers, customers, and stakeholders to resolve escalations and improve operational efficiency.
- Designed and optimized SQL database schemas, reporting systems, dashboards, and business intelligence modules providing visibility into fleet utilization, shipment status, delivery performance, and logistics KPIs.
- Collaborated with business stakeholders to translate operational requirements into scalable technology solutions.
- **Technologies:** Python, SQL, REST APIs, HTML, CSS, JavaScript, Transportation Operations, ELD Compliance, Fleet Compliance, Reporting & Analytics

SOFTWARE ENGINEER III *Movius Corporation*

March 2021 – July 2023

Product: MultiLine Desktop (T-Mobile)

Alpharetta, GA, USA

- Served as a Full Stack Developer and Product Architecture contributor for a secure enterprise communication platform supporting business messaging and calling.
- Participated in product architecture discussions and system design initiatives focused on scalability, maintainability, performance, observability, and platform reliability.
- Designed and maintained scalable front-end architecture using Angular, RxJS, NgRx, and Angular Router.
- Developed backend services, REST APIs, microservices, and SQL integrations supporting enterprise communication workflows.
- Led development of key product features including Voicemail, Group Messaging, and Picture Messaging (MMS).
- Integrated SIP protocols, WebSockets, and event-driven services supporting real-time communication capabilities.
- Worked with real-time communication systems, messaging infrastructure, and device-connected enterprise applications requiring high availability and low-latency data exchange.
- Developed a strong understanding of communication protocols, distributed architectures, connected ecosystems, and data flows commonly used in IoT environments.
- Collaborated across engineering, product, and customer-facing teams to translate business requirements into scalable platform capabilities.
- Built and optimized Electron-based desktop functionality across multiple operating systems.
- **Technologies:** Angular, TypeScript, Electron, Node.js, Java, SQL, SIP, WebSockets, RxJS, NgRx, Distributed Systems

SOFTWARE ENGINEER

August 2020 – February 2021

- Designed and developed responsive, mobile-first web applications using Angular, React, TypeScript, JavaScript, HTML5, CSS3, and SCSS.
- Built customer-facing e-commerce experiences supporting device selection, service plans, checkout workflows, and payment processing.
- Developed reusable UI components and integrated REST APIs supporting scalable business operations.
- Developed and integrated communication services supporting connected mobile ecosystems, real-time data exchange, and network-enabled applications.
- Gained experience working with distributed systems, API integrations, communication technologies, and connected device ecosystems commonly found in IoT environments.
- Implemented accessibility (WCAG), localization (i18n), and performance optimization strategies.
- Collaborated with Agile teams to deliver high-quality customer-facing software solutions.
- **Technologies:** Angular, React, TypeScript, JavaScript, HTML5, CSS3, SCSS, REST APIs, Connected Systems, Distributed Architectures

SOFTWARE ENGINEER / BUSINESS DEVELOPMENT MANAGER

August 2015 – July 2020

VirtuLive Technologies Pvt. Ltd.

Mysore, Karnataka, India

Product: Roombr Video Conferencing Platform

- Generated approximately INR 10 Crore (~\$1.2M USD) in revenue through strategic business development and enterprise software adoption initiatives.
- Designed and developed the Roombr Video Conferencing Platform supporting enterprise collaboration, virtual meetings, online classrooms, and digital communication services.
- Built scalable frontend applications using Angular, TypeScript, JavaScript, HTML5, CSS3, SCSS, and Electron.
- Developed backend services, REST APIs, and microservices using Node.js and Spring Boot.
- Engineered real-time communication features using WebRTC, SIP protocols, and WebSockets.
- Worked directly with enterprise customers, educational institutions, and government stakeholders to drive platform adoption and business growth.
- Participated in solution architecture discussions, product planning, customer demonstrations, and technical consulting activities.
- **Technologies:** Angular, TypeScript, Electron, Node.js, Spring Boot, SQL, MySQL, WebRTC, SIP, WebSockets

PROJECTS

Tale of Treasures – AI Storytelling Platform | Personal / Graduate Capstone Project | GitHub Available

- Designed and developed an AI-powered storytelling platform that generates interactive and personalized stories using Large Language Models (LLMs).
- Implemented Retrieval-Augmented Generation (RAG) using OpenAI APIs, ChromaDB, FAISS, embeddings, and vector search techniques to improve contextual story generation and knowledge retrieval.
- Built intelligent conversational workflows that dynamically generate stories based on user prompts, narrative context, and storytelling progression.
- Applied prompt engineering techniques to improve story quality, consistency, user engagement, and response relevance.
- Developed backend services, APIs, and application logic supporting AI-driven content generation and conversational interactions.
- Independently managed architecture, development, testing, deployment, and continuous enhancement of the platform as part of my graduate capstone project.
- Source code and project demonstrations available on GitHub.
- **Technologies:** Python, OpenAI API, ChromaDB, FAISS, Embeddings, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLMs, Generative AI, AI Agents, REST APIs

TECHNICAL SKILLS

Programming & Frameworks: JavaScript, TypeScript, Java, Python, Angular, React, Node.js, Spring Boot, HTML5, CSS3, SCSS, RxJS, NgRx, Redux

AI, Data & Analytics: Generative AI, LLMs, OpenAI API, Retrieval-Augmented Generation (RAG), ChromaDB, FAISS, Embeddings, Prompt Engineering, AI Agents, Vector Search, Business Intelligence, Reporting & Analytics, Operational Dashboards, KPI Monitoring, Trend Analysis

Backend, Database & Integration: REST APIs, Microservices, API Integration, SQL, MySQL, Database Design, Query Optimization

Cloud, DevOps & Architecture: Docker, Git, Azure DevOps, Jenkins, GitHub Actions, CI/CD, System Design, Product Architecture, Application Architecture, Microservices Architecture, Scalability Engineering, Design Patterns, Performance Optimization

KEY EXPERTISE

Full Stack Development, Product Architecture, System Design, Transportation Management Systems (TMS), Fleet Telematics Platforms, Enterprise Communication Platforms, Real-Time Messaging Systems, IoT Solutions, Device Telemetry, ELD Compliance Workflows, Fleet Compliance, REST API Development, Microservices Architecture, Database Design & Optimization, Technical Leadership, Agile Software Development